

DFT Validation of Ground-Truth Magnetic Ordering Labels

24-candidate real-DFT check against Materials Project's stored magnetic ordering

Prepared for advisor review · 21 August 2026 · snapshot, 2 of 26 still running

Headline result

Of 24 shortlisted candidates checked so far against real VASP DFT (26 eligible, 2 still running), **4 are confirmed** with a resolvable margin, **17 are clearly overturned** to AFM (0.31–24.68 meV/atom more stable than FM), and **3 are statistical ties** (≤ 0.10 meV/atom, inside VASP's own convergence tolerance).

1. Purpose

This project's screening pipeline treats Materials Project's stored magnetic ordering as ground truth (`confidence=1.0`) wherever it is the source database. A first real-DFT check on one shortlisted candidate (mp-861559, TaFeO4) found that label overturned — MP's own workflow relaxes each structure from a single initial magnetic-moment guess and never enumerates AFM alternatives, so its "ground truth" reflects whatever configuration that guess happened to converge to. A follow-up 6-candidate batch confirmed this wasn't a one-off (0 of 6 confirmed). This report extends the check to the full set of eligible `ground_truth / FiM` candidates in the shortlist — 24 of 26 complete so far.

2. Method

Identical pipeline for every candidate, matching Materials Project's own magnetic-ordering convention so results are directly comparable to MP's stored values:

1. Take the candidate's relaxed structure from the harmonized dataset.
2. Enumerate symmetry-distinct FM/AFM spin configurations with pymatgen's `MagneticStructureEnumerator` (backed by `enumlib`). The enumerator only ever produces FM/AFM configurations — never FiM, even for the 2 candidates here labeled `FiM` (mp-22972, mp-551086). For those two, the result only rules out FM; it does not confirm or refute the FiM label itself.
3. Generate VASP inputs with `MPRelaxSet` — MP's own GGA+U, ENCUT, and k-point density — overriding only `MAGMOM` per ordering.
4. Run each configuration to convergence (48 cores/job). The original 6 candidates ran on `paramsanganak` (VASP 5.4.4); all later candidates ran on `paramrudra` (VASP 6.4.3) after `paramsanganak` became unavailable mid-project.
5. Compare converged total energies **per atom** (not raw `TOTEN`); the lowest is the DFT ground state.

All configurations for the 24 complete candidates converged cleanly — every `OUTCAR` reports "reached required accuracy." 2 candidates are still running: mp-1173143 (1 of 4 configs converged) and mp-682554 (1 of 4 configs converged, a sulfide with no LDA+U applied — a distinct, unrelated failure mode, not part of the tungsten issue below).

Data-integrity note

Two bugs in the analysis script were found and fixed while processing this expanded batch: (1) a ground-state comparison that used raw `TOTEN` instead of energy per atom — since AFM configurations are enumerated in supercells 2–4× the FM primitive cell, this systematically favored whichever configuration had more atoms, independent of actual stability; (2) non-converged, TIME-LIMIT-truncated energies were briefly included in ground-state comparisons. Both are fixed. The original 6-candidate numbers below are unaffected — verified unchanged after the fix — but the bug would have silently corrupted several of the newly completed candidates had it not been caught.

Hubbard U on tungsten removed for 5 candidates (methodological deviation)

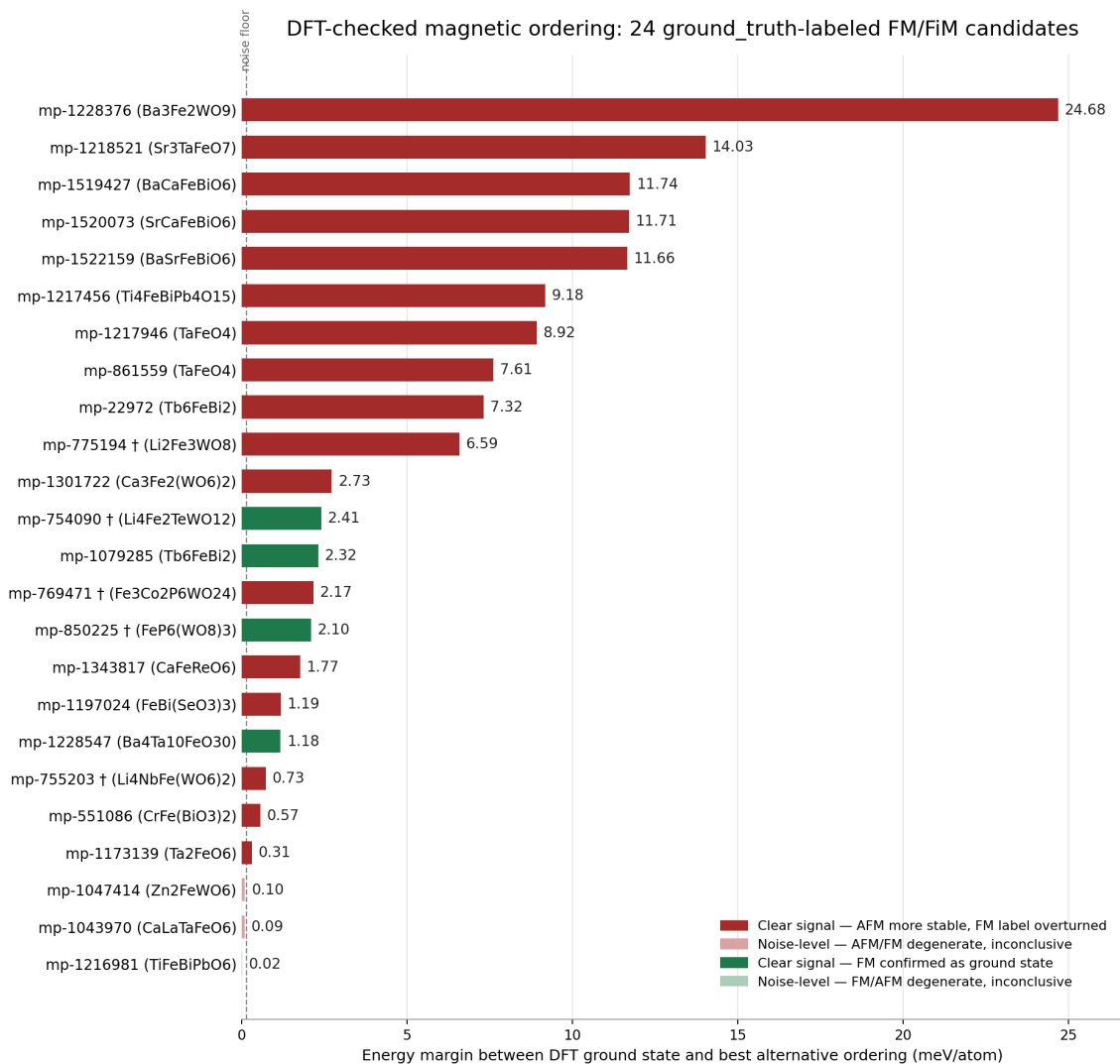
All 5 W-bearing candidates (mp-754090, mp-755203, mp-769471, mp-775194, mp-850225) initially failed to converge under MP's standard GGA+U value ($U=6.2$ eV on W). Root cause: W in these compounds is formally W^{6+} , i.e. nominally d^0 — a large U on a near-empty d-shell makes the +U energy almost invariant to how that near-zero occupancy distributes across W's five d orbitals and across inequivalent W sites, so the SCF charge density has no restoring force and oscillates indefinitely rather than converging. Confirmed with 11 independent sandbox tests (three SCF algorithms, two mixing schemes, an `LMAXMIX` control, and a warm start from a converged no-U density) — none converge with $U=6.2$ kept on W. The only working fix is removing U from W entirely (`LDAUL=-1` on the W species), applied uniformly across all orderings of a given compound so the FM/AFM energy comparison stays valid. This reproduced to within 0.5 meV/atom across two independent mixers in the sandbox, and converged all 18 production orderings cleanly on the first attempt (checked against false convergence via `rms(c)`, not just the "EDIFF is reached" banner). **Consequence:** the reported energies for these 5 candidates are not on the same U-value footing as MP's own database values or the other 19 candidates in this report, so cross-candidate energy comparisons involving these 5 should be treated with that caveat in mind; the within-compound FM-vs-AFM verdict for each is unaffected, since the same U treatment applies to every ordering of a given compound.

3. Results

24 candidates complete, sorted by margin between the DFT ground state and the best alternative ordering. A margin below ~ 0.15 meV/atom is inside typical VASP electronic-convergence tolerance (EDIFF= $1e-6$ eV here) and is read as a statistical tie. The 5 rows marked † are the tungsten candidates fixed this snapshot (U removed from W — see the data-integrity note above).

Candidate	Formula	Label	DFT ground state	Margin (meV/atom)	Signal
mp-1228376	Ba3Fe2WO9	FM	AFM	24.68	clear — overturned
mp-1218521	Sr3TaFeO7	FM	AFM	14.03	clear — overturned
mp-1519427	BaCaFeBiO6	FM	AFM	11.74	clear — overturned
mp-1520073	SrCaFeBiO6	FM	AFM	11.71	clear — overturned
mp-1522159	BaSrFeBiO6	FM	AFM	11.66	clear — overturned
mp-1217456	Ti4FeBiPb4O15	FM	AFM	9.18	clear — overturned
mp-1217946	TaFeO4 (entry 2)	FM	AFM	8.92	clear — overturned
mp-861559	TaFeO4 (entry 1)	FM	AFM	7.61	clear — overturned
mp-22972	Tb6FeBi2	FiM	AFM	7.32	clear vs FM, doesn't test FiM
mp-775194 †	Li2Fe3WO8	FM	AFM	6.59	clear — overturned
mp-1301722	Ca3Fe2(WO6)2	FM	AFM	2.73	clear — overturned
mp-754090 †	Li4Fe2TeWO12	FM	FM	2.41	clear — confirmed
mp-1079285	Tb6FeBi2 (entry 2)	FM	FM	2.32	clear — confirmed
mp-769471 †	Fe3Co2P6WO24	FM	AFM	2.17	clear — overturned
mp-850225 †	FeP6(WO8)3	FM	FM	2.10	clear — confirmed
mp-1343817	CaFeReO6	FM	AFM	1.77	clear — overturned
mp-1197024	FeBi(SeO3)3	FM	AFM	1.19	clear — overturned
mp-1228547	Ba4Ta10FeO30	FM	FM	1.18	clear — confirmed
mp-755203 †	Li4NbFe(WO6)2	FM	AFM	0.73	clear — overturned
mp-551086	CrFe(BiO3)2	FiM	AFM	0.57	

					clear vs FM, doesn't test FiM
mp-1173139	Ta ₂ FeO ₆	FM	AFM	0.31	clear — overturned
mp-1047414	Zn ₂ FeWO ₆	FM	AFM	0.10	noise-level
mp-1043970	CaLaTaFeO ₆	FM	AFM	0.09	noise-level
mp-1216981	TiFeBiPbO ₆	FM	FM	0.02	noise-level



Breakdown of the 24

17 clearly overturned to AFM (0.31–24.68 meV/atom); **4 clearly confirmed** as FM (mp-1079285, mp-1228547, mp-754090 †, mp-850225 †; 1.18–2.41 meV/atom); **3 noise-level ties** (≤ 0.10 meV/atom: mp-1047414, mp-1043970, mp-1216981) that cannot be called either way. Among the 21 that resolve clearly, "AFM overturns FM" outnumbered "FM confirmed" 17 to 4.

4. Conclusion

Extending from the original 6-candidate check (0 of 6 confirmed) to the full 24-candidate batch changes the picture only modestly: **4 of 24 are now confirmed** with a real margin (all in the newer paramrudra batch), but **17 of 24 remain clearly overturned**, reinforcing rather than undercutting the original finding. The mislabeling is not universal — a small minority of `ground_truth` FM labels do hold up under real DFT — but the majority in this shortlisted population do not. This is consistent with MP's known workflow gap: relaxation from one initial magnetic-moment guess, with no systematic AFM enumeration, means MP's stored ordering reflects whichever configuration that one guess happened to converge to, which is right often enough to not be pure noise, but wrong on a majority of structures in this DMI-filtered population.

On the FiM-labeled candidates: mp-22972 and mp-551086 are labeled `FiM`, not `FM`, but the enumerator used here never generates FiM configurations. AFM beating the FM baseline for these two does not confirm or refute their actual FiM label — it only rules out FM.

Caveat on sample composition

This is still not a random sample — all candidates were pulled from the top of the DMI-filtered shortlist (noncentrosymmetric, heavy-SOC-element, net-moment structures). It is plausible that this structural family is more AFM-prone than the `ground_truth` population as a whole. What this sample does establish firmly: the label cannot be taken at face value for candidates in this specific shortlisted population without an actual DFT check.

5. Recommendation for discussion

1. **Headline** (24-candidate snapshot, 2 remaining): of 24 shortlisted `ground_truth` / `FiM` labels checked against real DFT, 17 are clearly wrong, 4 are confirmed with a resolvable margin, and 3 are statistical ties. The confirmation rate with a real margin is roughly 1 in 6.
2. **Implication:** the classifier's/MP's ordering label should be treated as a screening prior, not ground truth, for this shortlist's structural family. Worth deciding whether to re-flag affected shortlist entries or revisit how `ordering_final` feeds the net-moment filter in `screen_candidates.py` (AFM is currently excluded from the DMI funnel entirely).
3. **Open question:** is 24–26 of 1,647 candidates enough evidence to act on, or does this need a larger, more representative validation batch — e.g. a random sample stratified across the whole `ground_truth` population, not just the DMI-filtered top of the shortlist — before changing how the pipeline treats these labels?
4. **Follow-up worth scoping:** a dedicated FiM-configuration check for mp-22972 and mp-551086, since FM/AFM enumeration structurally cannot confirm or refute an FiM label.

Raw data

- Per-configuration results: `outputs/dft_ordering_validation.csv` (all candidates with any converged data; `candidate_complete` marks which of the 24 are final)

- Staged calculations: `data/dft_staging/<candidate_id>/{0_fm,1_afm,...}/` (paramrudra runs write directly into the calc dir; paramsanganak runs nest under a `results/` subdirectory)
- Full writeup: `docs/dft_validation_ground_truth_labels.md`
- Candidate labels: `outputs/phase4_shortlist.csv`